

Mean Decrease Accuracy (MDA)

MDA is also often referred as Permutation Feature Importance.

Unlike MDA, it evaluates how model performs on OOS data (out of sample)

How the "shuffling" (Permutation) works?

1. Train and score: accuracy with training data becomes baseline performance (e.g. Accuracy = 85%)
2. Shuffle one column's rows: breaks the link between feature and target. If feature is important should impact model accuracy.
3. Re-score: measure new performance (e.g. Accuracy = 60%)
4. Calculate importance: the difference between baseline and new (Importance = $85 - 60 = 25\%$ drop)

Critical Considerations

- Model agnostic: just needs to run predictions on model
- Metric flexibility: can use F1-score, Negative log-loss...
- Substitution effect: same as MDI, there is a collinearity effect.
- Realistic importance: possible for a feature to have zero, even negative importance
- Need for Purged and Embargoed CV: in time-series data, training set information could leak in validation set, making MDA scores optimistic.